

Claim Denial Risk Prediction System

1. Executive Summary

This project delivers a production-ready pre-bill denial prediction system that combines classical machine learning with a GenAI explanation engine. The system scores 500 current claims for denial risk, ranks them by priority, and generates plain-English corrective action recommendations for the highest-risk claims.

After evaluating 10 model variants across 5 algorithm families, a Calibrated Logistic Regression model was selected for production deployment. It captures **45.7% of all denials within the top 25% review window** -- nearly double the 25% expected from random review -- and provides reliable probability estimates with a Brier score of 0.137.

2. Business Impact

Denied claims are one of the largest sources of revenue leakage in healthcare revenue cycle management. Each denied claim requires rework -- investigation, correction, resubmission, and follow-up -- costing \$25-118 per claim in administrative overhead. Pre-bill denial prediction moves this intervention upstream: instead of reacting to denials after submission, billers can fix issues before the claim leaves the door.

- 45.7% of denials captured in the top 25% highest-risk claims -- nearly 2x random review
- 125 claims flagged as High risk for immediate biller attention
- All 125 High-risk claims receive AI-generated plain-English explanations with specific corrective actions

3. Key Findings

The most predictive risk factors are administrative completeness checks: whether the claim has a required prior authorization, whether supporting documentation is attached, whether a required referral is present, and whether the provider is in-network. These are all actionable before submission, making the system operationally practical.

Ten model architectures were evaluated. Logistic Regression variants achieved the highest performance (49.51% capture on validation), while Gradient Boosting was the best non-linear alternative at 48.54%. The strong performance of linear models confirms that administrative gap flags have a direct, additive relationship with denial outcomes.

4. Model Selection Results

The table below shows the top 5 models ranked by denial capture rate within the top 25% review window. The Calibrated Logistic Regression model was selected for production because it matches the best capture rate while providing substantially more reliable probability estimates (34% improvement in calibration).

Model	Type	Capture@25%	ROC-AUC	Brier
Calibrated Logistic Regression	Linear	49.51%	0.710	0.137 -- Best calibration
Logistic Regression Baseline	Linear	49.51%	0.711	0.209
Gradient Boosting (GBM)	Tree Ensemble	48.54%	0.708	0.165 -- Best non-linear
Voting Ensemble (LR+RF+GBM)	Ensemble	48.54%	0.705	0.172
Random Forest	Tree Ensemble	39.81%	0.653	0.155

Final Test Set Performance

Metric	Value
Model Deployed	Calibrated Logistic Regression
Denial Capture at 25% Review	45.7%
Precision at 25% Review	47.8%
ROC-AUC	0.691
Brier Score (Calibrated)	0.137
High-Risk Claims Flagged	125 of 500
LLM Model	gemma4:31b-cloud (Ollama)

5. GenAI Explanation Engine

The system uses Ollama Cloud's gemma4:31b-cloud model to generate plain-English explanations for all 125 High-risk claims, with deterministic templates covering the remaining 375 Medium and Low-risk claims. Each explanation contains three elements:

- A statistical disclaimer ("This is a statistical estimate and not a guaranteed outcome")
- Identification of specific risk factors (missing authorization, missing documentation)
- Actionable corrective steps (resolve, attach, verify, confirm)

Every AI explanation is validated through a Pydantic schema before being accepted. The schema enforces disclaimer presence, minimum length, and structural completeness. If validation fails, a deterministic template is used as fallback. In the latest production run: 125 API calls (all High-tier), 375 deterministic templates (Medium/Low), 100% validation pass rate.

6. Guardrails

The system includes a production-grade LLM audit infrastructure that records every API call with full metadata: token usage (prompt + completion), latency, JSON parse success,

Pydantic validation success, and quality flags (disclaimer presence, actionable content, hallucination markers). No patient data is sent to the LLM -- only engineered features and risk factor labels. All audit logs are stored locally in JSON format with no external telemetry.

Capability	Details
Token Tracking	Prompt tokens + completion tokens from Ollama ChatResponse
Latency Logging	Total duration and evaluation duration per call
Response Validation	JSON parse success, Pydantic schema validation
Quality Checks	Disclaimer, action verbs, min length, PII/hallucination scan
Storage	Local JSON audit logs in data/output/audit_logs/
HIPAA Alignment	No external telemetry, no PII in prompts, response truncation